# Written Discussion (roughly one page, as required)

Comparison tables and histograms are in results_1_2.pdf and figures/; this page is the written discussion only.

**Soundness versus pessimism (Section 1.7).** Measurement-based estimation is cheap and tight for paths actually exercised, but unsound: it reports only the slowest observed run, never a guarantee. Static analysis is sound but often pessimistic. This tool sits on the unsafe side, then adds a margin (+54% from a 65% utilization budget) as a heuristic buffer, not a proof.

**Where the tool was confident.** For fir_filter (fixed control flow), the random and adversarial strategy means are nearly equal (0.070 ms vs 0.072 ms), so its high max/p99 ratios reflect timing noise, not a data-dependent path — the tool is reasonably confident here.

**Where it likely underestimated.** For the three data-dependent functions, random testing alone underestimates the true worst case badly compared with a hand-derived adversarial input: early_exit_search 3.10x, classify_batch 3.41x, collatz_steps 2.41x. A tool given no knowledge of the function's internals would miss these heavier paths, since random inputs rarely reach the deepest branch or longest chain.

**Certification caveat (Section 1.8).** A sample of 1000 calls gives no path-coverage guarantee, and interpreter/OS noise (garbage collection, scheduling, frequency scaling) cannot be bounded this way. The margin does not repair this: for every function tested, random-only-max x (1 + margin) still falls short of the true adversarial maximum. Certification needs an argued, sound bound — exhaustive path analysis or a hand-verified worst-case input — which this measurement harness alone cannot supply.